MILESTONE 4: TEAM „WHALE“

Members: Qianyue Li, Zheng Lin, Yanhong Xu - **Dataset:** Reward Task

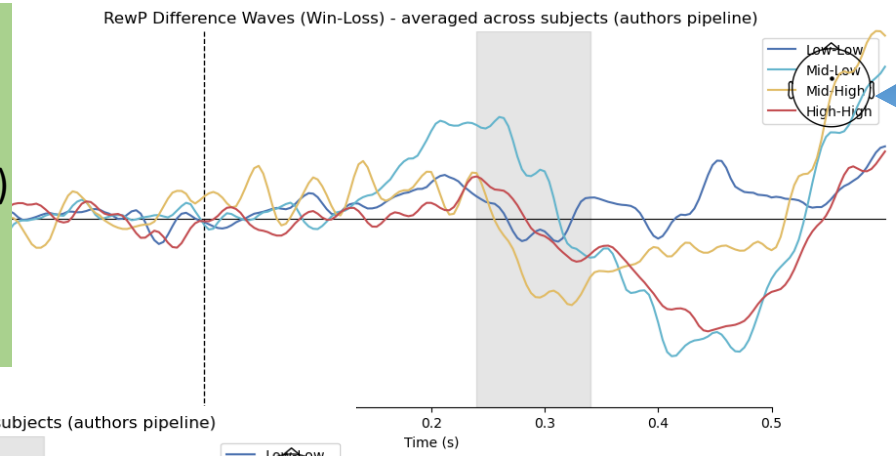
1. Main Research Question

Whether and how reward-related anterior cingulate cortex (ACC) activity, as indexed by reward positivity (RewP) varies with task value.

Our own pipeline

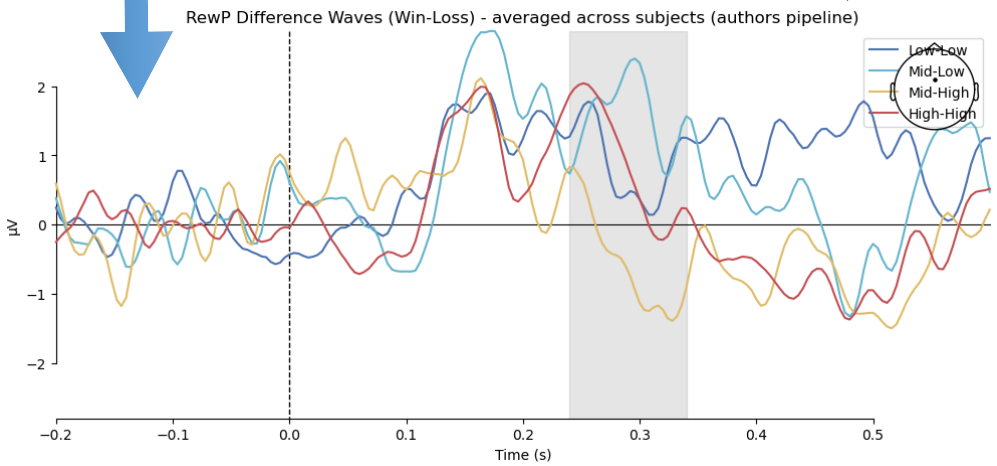
Difference from author's pipeline

- trial rejection criteria
- Ica method (Picard instead of Infomax)
- Ic rejection criteria
- Only participants from one cite
- Trimmed mean

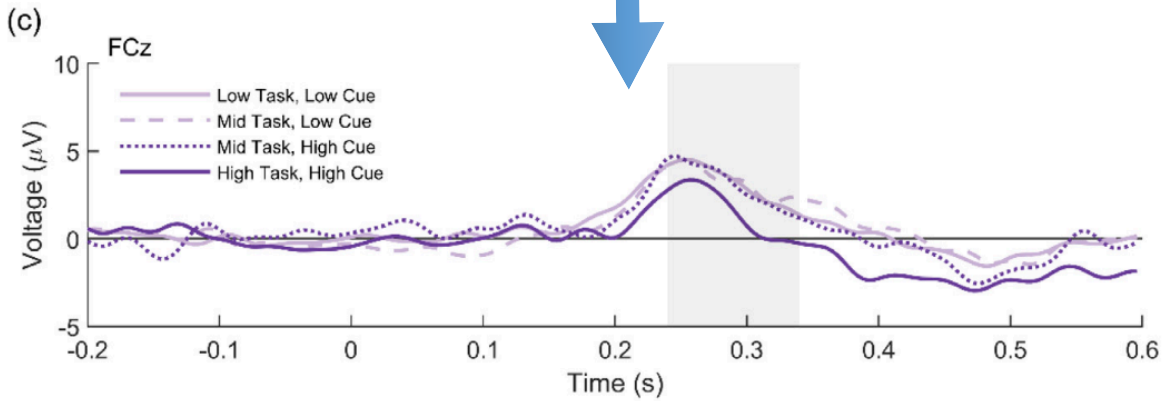


Us following the authors' pipeline (roughly)

- Same preprocessing
- Same trial rejection & ica , etc.
- Only participants from one cite
- Trimmed mean

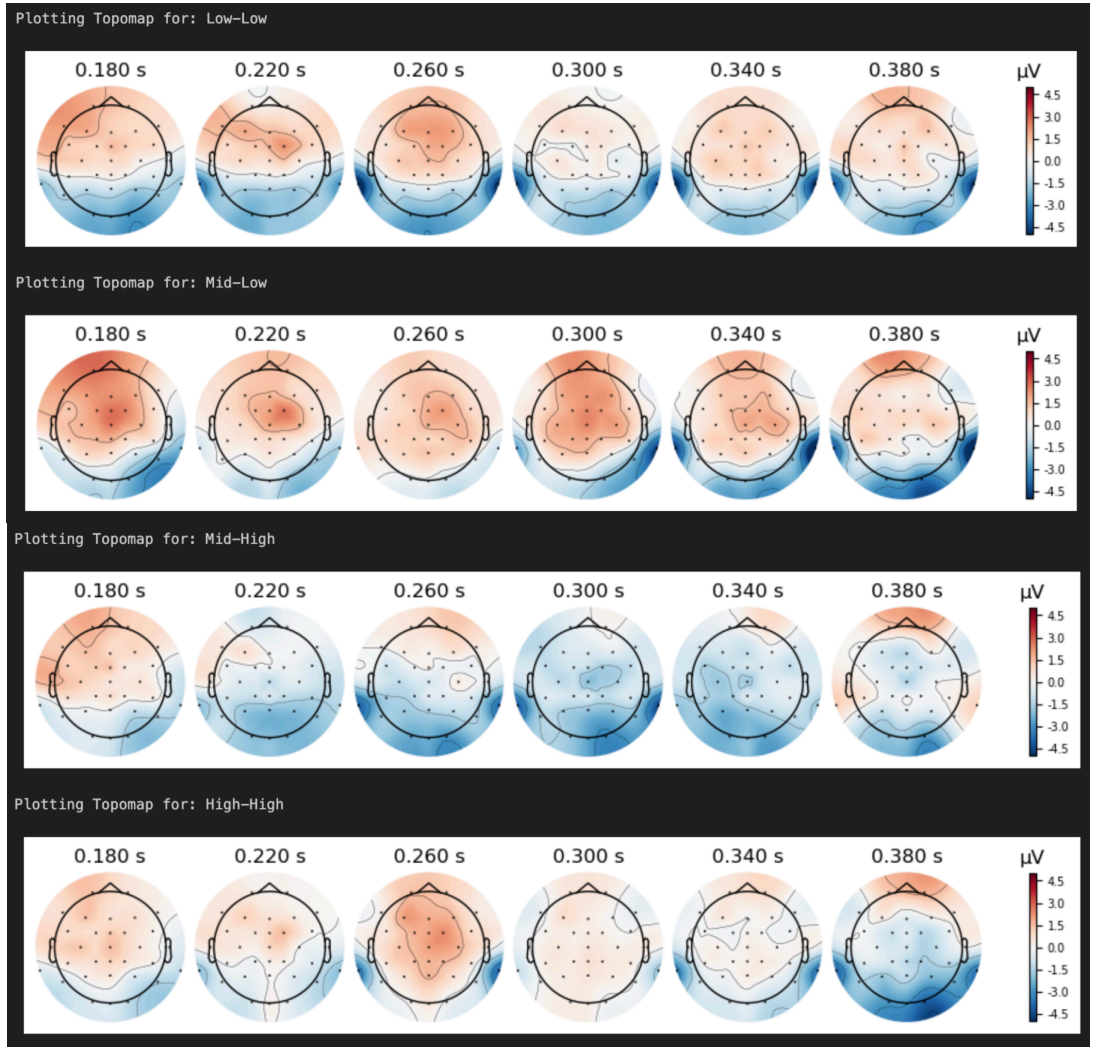


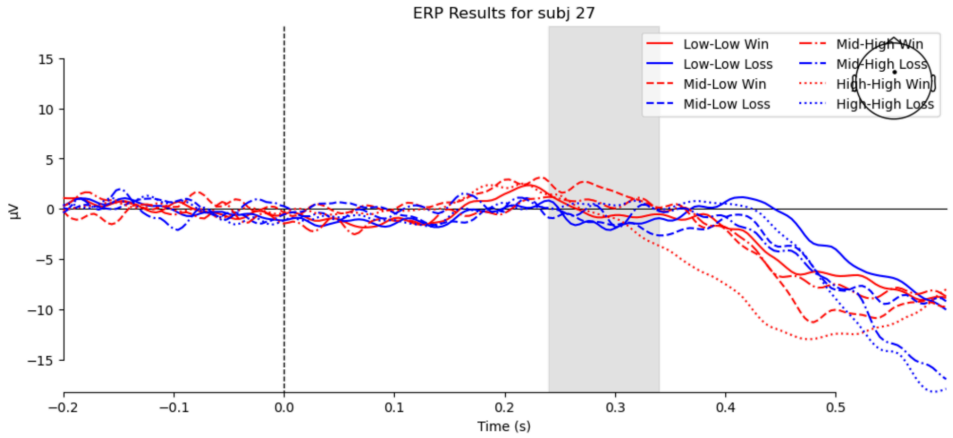
Authors' results



3. The problems...

Topography series of the average for different conditions





Before: IC rejection criteria: eye blink > 50% - too low?

After: IC rejection criteria: brain < 40% - too high?

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Transforming to ICA space (31 components)
Zeroing out 18 ICA components
Projecting back using 32 PCA components
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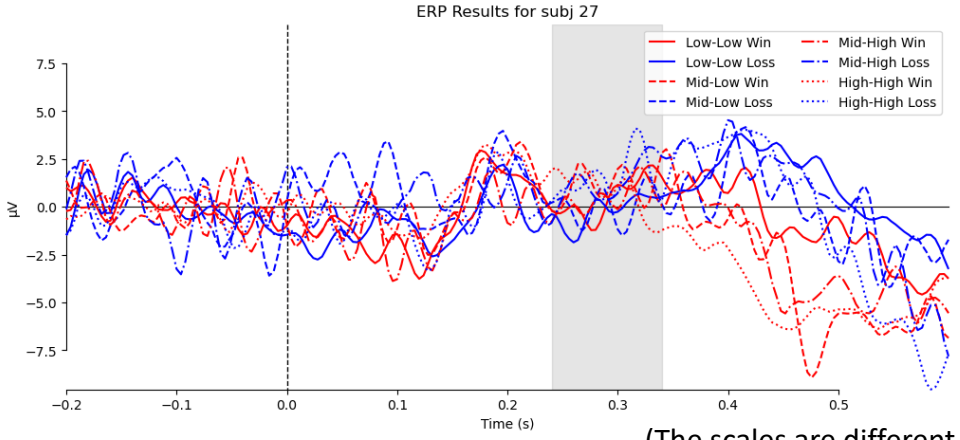
Should we use a uniform criteria for all subjects or inspect individual ones and make different criterion based on the observations?



(The scales are different but still)